

VIDYATRACK - QA TEST SCENARIOS

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VIDYATRACK – CROSS-CUTTING QA TEST SCENARIOS Applies Across Roles: TEACHER / PRINCIPAL / MANAGEMENT

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1) Cross-Role Boundary Testing (P0)

Purpose:

Ensure role transitions and boundary conditions do not allow unauthorized access or stale permissions.

SCENARIOS:

- Teacher becomes INACTIVE → existing session attempts:

- POST /attendance
- POST /marks/record

Expected: 403 immediately (no grace period)

- Teacher replaced → old teacher:

- Cannot view section students
- Cannot mark attendance or marks

Expected: access revoked instantly

- Principal attempts:

- POST /attendance
- POST /marks/submit
- POST /students

Expected: 403 (backend enforced)

- Management attempts teacher-only endpoints:

- /teachers/me/*

Expected: 403/404

PASS CRITERIA:

- No role can perform actions outside its permission set
 - Backend blocks all unauthorized access even with valid token
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2) Cross-Academic-Year Scenarios (P0 / P1)

Purpose:

Ensure strict isolation of academic data by academic year.

SCENARIOS:

- Create new academic year
- Verify:
 - Attendance for previous year not editable
 - Marks from previous year not overwritten
- Promote students to new class/section
- Teacher views:
 - Only current year students for attendance
- Management views:
 - Historical data remains read-only

NEGATIVE TESTS:

- Attempt attendance submission using old academic year context

Expected:

- Blocked or stored strictly under correct year

PASS CRITERIA:

- No data bleed between academic years
- Historical records remain intact and immutable

3) Multi-Section / Multi-Assignment Teachers (P0)

Purpose:

Validate teacher scope when assigned to multiple sections/subjects.

SCENARIOS:

- Teacher assigned to:
 - Section A – Subject X
 - Section B – Subject Y
- Verify:
 - Attendance allowed only for attendance-section endpoint result
 - Marks allowed only for assigned (section, subject) pairs

NEGATIVE TESTS:

- Teacher attempts marks for:
 - Assigned section but unassigned subject
 - Unassigned section

Expected:

- 403/404

PASS CRITERIA:

- Fine-grained enforcement per assignment
- No over-permission leakage

4) Notification Idempotency & Ordering (P0)

Purpose:

Prevent duplicate, missing, or misordered parent notifications.

SCENARIOS:

- Trigger attendance → notification queued
- Re-run outbox worker multiple times
- Simulate:
 - WhatsApp failure → SMS fallback
 - Worker crash mid-run → re-run

NEGATIVE TESTS:

- Same attendance event processed twice
- Expected:
- Only one notification sent per student per event

- Out-of-order processing

Expected:

- No duplicate or contradictory messages

PASS CRITERIA:

- Notifications are idempotent
- Delivery status accurately reflects final state
- No wrong-recipient or duplicate messages

5) Audit & Forensics Testing (P0)

Purpose:

Ensure all critical actions are traceable for dispute resolution.

SCENARIOS:

Verify audit evidence exists for:

- Attendance marked (who, when)
- Attendance edited (same-day)
- Marks recorded

- Marks submitted
- Student notes added
- Parent messages sent

NEGATIVE TESTS:

- Action performed → no audit trace
- Expected:
- FAIL (pilot blocker)

PASS CRITERIA:

- Every write action has:
 - actor_user_id
 - timestamp
 - entity reference
- Management/Principal can view audit data (read-only)

6) Data Deletion / Orphan Safety (P0)

Purpose:

Prevent orphaned or corrupted academic data.

SCENARIOS:

Attempt to delete:

- Class with sections
- Section with students
- Subject with marks
- Teacher with assignments

EXPECTED BEHAVIOR:

- Hard block OR safe failure with clear error
- No cascading deletes of academic data

NEGATIVE TESTS:

- Partial delete leaving orphan rows
- Expected:
- FAIL (data integrity breach)

PASS CRITERIA:

- Referential integrity preserved
- No orphan attendance, marks, notes, or assignments

7) Performance Sanity (Pilot Scale)

Purpose:

Ensure system remains usable at real pilot sizes.

SCENARIOS:

- Attendance submission with 60–80 students
- Marks submission with:
 - Multiple subjects
 - Full class
- Dashboard load with full data set

EXPECTED:

- No timeouts
- No server errors
- Acceptable response time (subjective, but consistent)

PASS CRITERIA:

- All actions complete successfully
- No partial writes or retries required due to slowness

8) Backup & Recovery Drill (P0 – Operational QA)

Purpose:

Validate operational safety before pilot.

SCENARIOS:

- Take full DB backup
- Perform destructive test:
 - Bad deployment OR
 - Intentional bad data insertion
- Restore from backup

EXPECTED:

- Data restored completely
- System operational post-restore

PASS CRITERIA:

- Backup exists and is usable
- Restore procedure documented and verified

9) Timezone & Date Boundary Tests (India Context) (P0)

Purpose:

Ensure date-based logic aligns with school timezone (IST).

SCENARIOS:

- Mark attendance at:
 - 11:59 PM IST
 - 12:01 AM IST
- Attempt same-day edit across boundary

NEGATIVE TESTS:

- Edit previous day attendance after midnight IST

Expected:

- Blocked

PASS CRITERIA:

- Date boundaries respect IST
- Same-day edit logic is correct
- No off-by-one-day errors in attendance or marks

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FINAL NOTE

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These scenarios are:

- Non-negotiable for pilot confidence
- High-risk, low-visibility failure points
- Typically missed without senior QA oversight

Completing these closes the remaining QA gaps for VidyaTrack.

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